

# CENG212-Devre Teorisi ve Uygulamaları

## H2CD1-Elektrik Devre Değişkenleri

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versiyon160226

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**Değerlendirme** : **Vize : %50**  
**Final: %50**

# Dersin Amacı

- Bu dersin amacı temel elektrik ve elektronik devre elemanlarını, bu elemanların karakteristiklerini tanıtmak, devre teoremlerini öğretmek ve bu teoremleri kullanarak öğrencilere bir devreyi analiz etme yeteneğini kazandırmaktır.

# Dersin İçeriği

- Elektrik devre değişkenleri
- Devre elemanları, aktif-pasif devre elemanları, bağımlı-bağımsız kaynaklar
- Dirençli devreler, Kirşof kanunları, akım ve gerilim bölücü devreler, devre analizi
- Dirençli devrelerin analiz metotları
- Devre Teoremleri
  - Düğüm gerilimi analizi,
  - Çevre akımları analizi
  - Kaynak dönüşümü,
  - Süperpozisyon,
  - Thevenin/Norton teoremleri,
  - Maksimum güç transferi

# Dersin İçeriği

- Enerji depolayan elemanlar: Kapasitörler, bobinler
- Elektroniğe giriş, yükselticiler, yükseltici devre modelleri, işlemsel yükselteçler- Op Amp, ideal Op Amp içeren devre analizleri
- Diyotlar, diyot Devrelerinin analizi, zener diyotlar, doğrultucu devreler, kırpıcı ve bağlayıcı devreler
- İki kutuplu jonksiyon transistörleri-BJT's, aktif moddaki npn ve pnp transistör işlemleri
- DC/AC ortamında transistörlü devre çözümleri
- Alan etkili transistörler-FET's
- DC ortamda alan etkili transistör devre çözümleri

# Kaynaklar

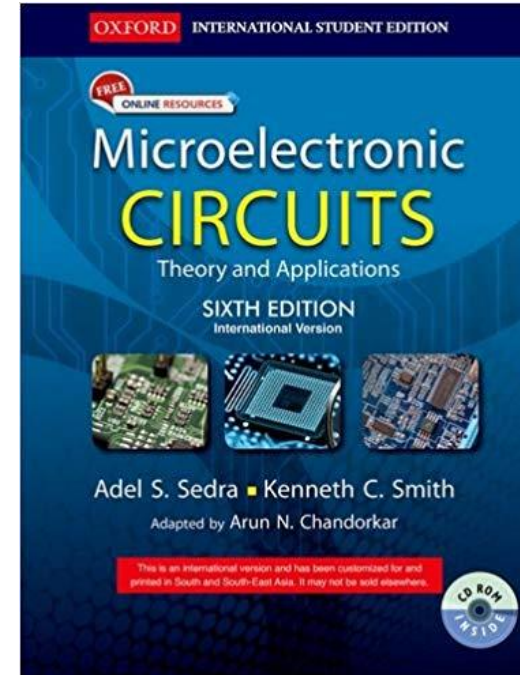
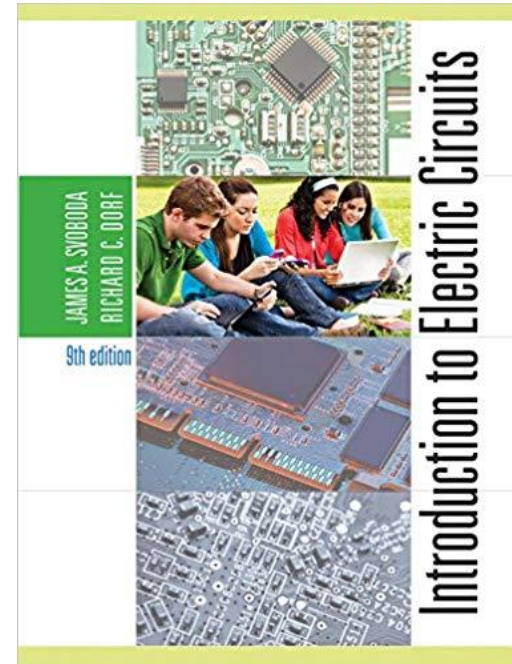
- Teori için

- “Introduction to Electric Circuits”, James Svoboda, Richard Dorf
- “Microelectronic Circuits”, Adel S. Sedra, Kenneth C. Smith

- Uygulamaya Meraklı Öğrenciler için

- “Arduino Robotics”, John D. Warren, Josh Adams, Harald Molle
- “Beginning Arduino”, Michael McRoberts
- “30 Arduino Projects for the Evil Genius”, Simon Monk

- EDS duyurularını takip ediniz.





JAMES A. SVOBODA  
RICHARD C. DORF

9th edition



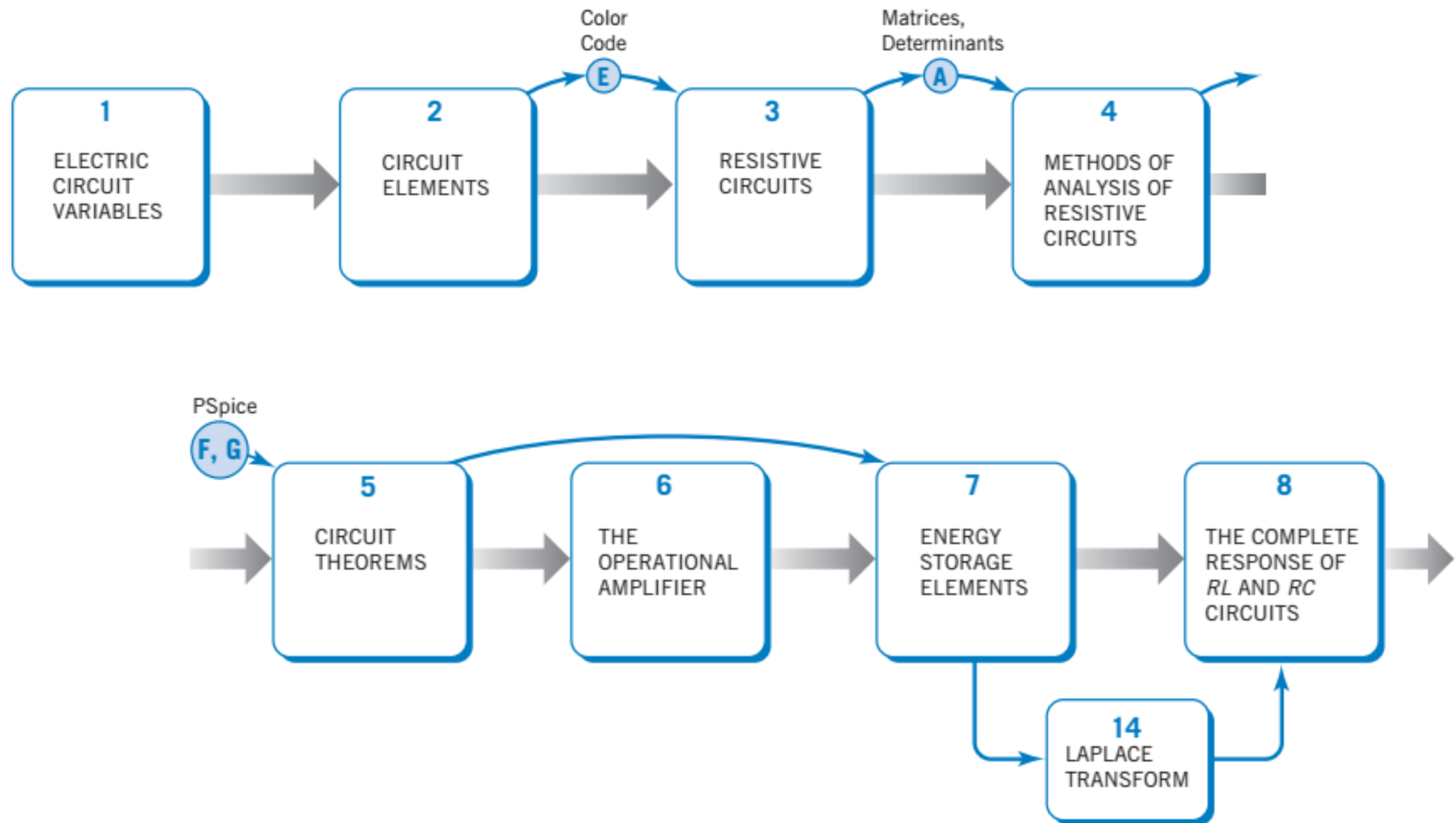
# Introduction to Electric Circuits

# SEDRA SMITH

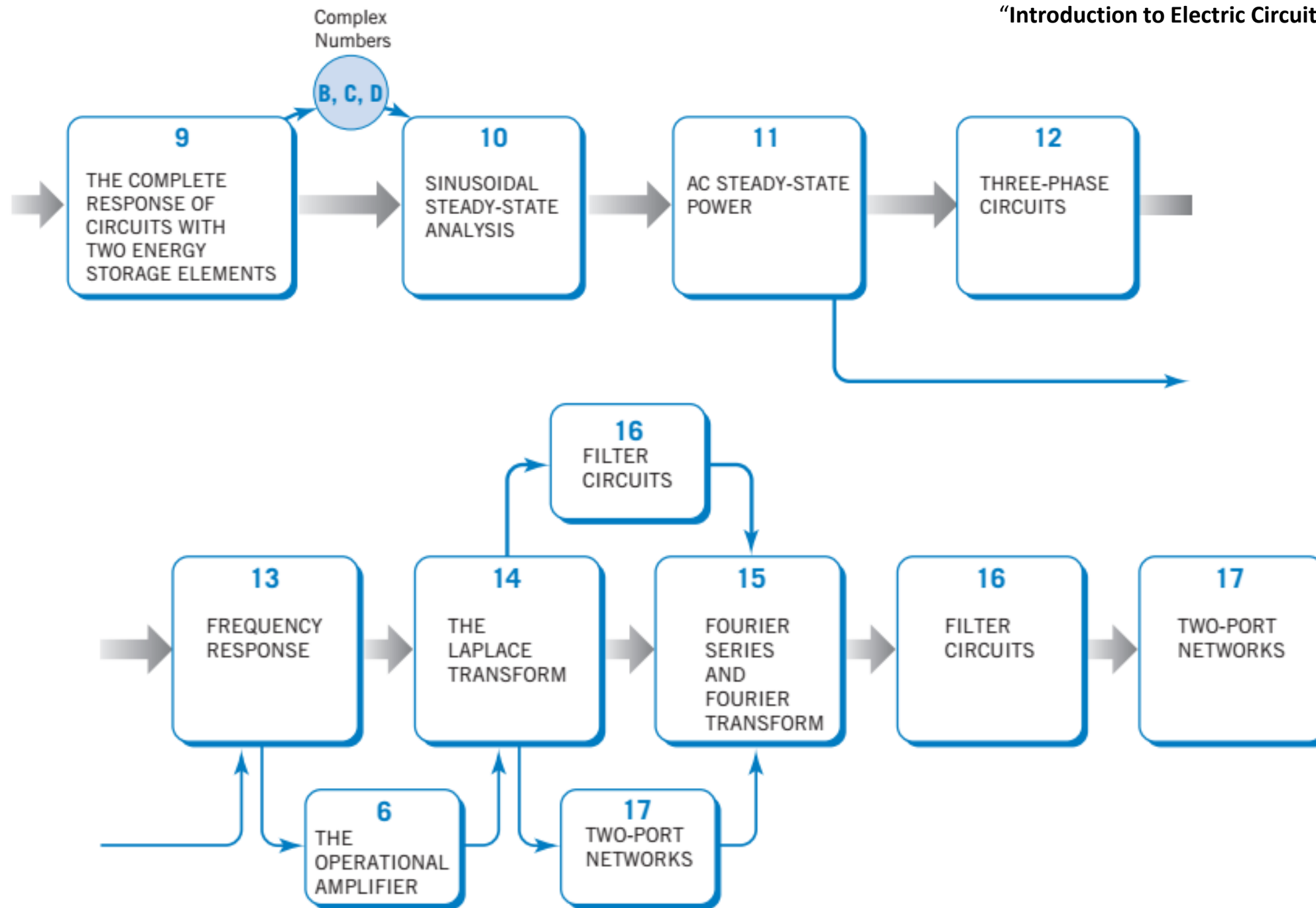
## MICROELECTRONIC CIRCUITS

INTERNATIONAL SEVENTH EDITION

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# CHAPTER 1



# *Electric Circuit Variables*

## IN THIS CHAPTER

**1.1** Introduction

**1.2** Electric Circuits  
and Current

**1.3** Systems of  
Units

**1.4** Voltage

**1.5** Power and  
Energy

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## 1.1 *Introduction*

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A circuit consists of electrical elements connected together. Engineers use electric circuits to solve problems that are important to modern society. In particular:

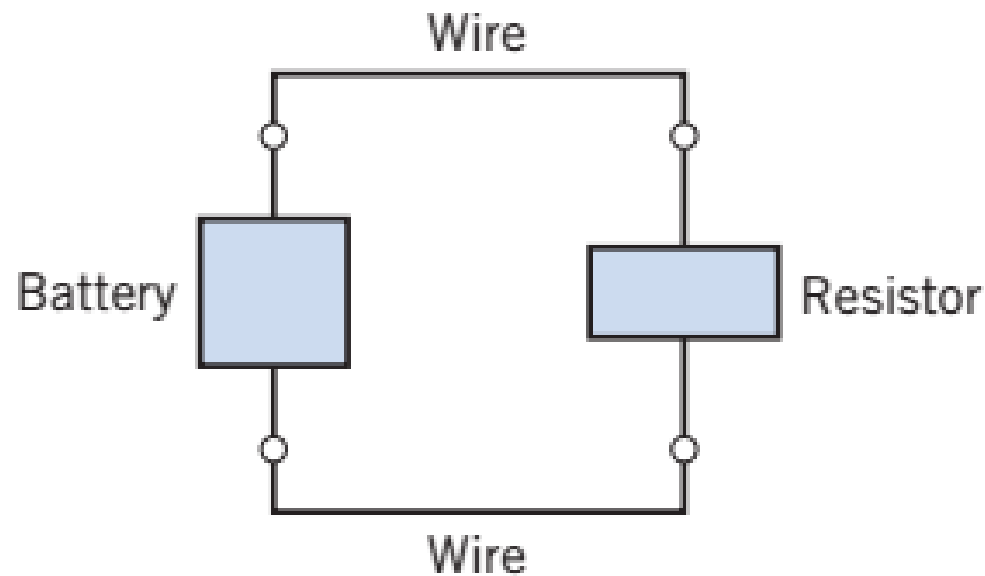
1. Electric circuits are used in the generation, transmission, and consumption of electric power and energy.
2. Electric circuits are used in the encoding, decoding, storage, retrieval, transmission, and processing of information.

In this chapter, we will do the following:

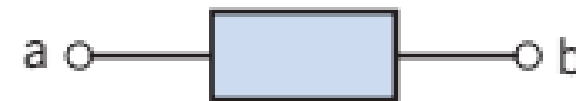
- Represent the current and voltage of an electric circuit element, paying particular attention to the reference direction of the current and to the reference direction or polarity of the voltage.
- Calculate the power and energy supplied or received by a circuit element.
- Use the passive convention to determine whether the product of the current and voltage of a circuit element is the power supplied by that element or the power received by the element.
- Use scientific notation to represent electrical quantities with a wide range of magnitudes.

## 1.2 Electric Circuits and Current

An **electric circuit** or electric network is an interconnection of electrical elements linked together in a closed path so that an electric current may flow continuously.



**FIGURE 1.2-1** A simple circuit.



**FIGURE 1.2-2** A general two-terminal electrical element with terminals a and b.

Charge may flow in an electric circuit. *Current is the time rate of change of charge past a given point.* Charge is the intrinsic property of matter responsible for electric phenomena. The quantity of charge  $q$  can be expressed in terms of the charge on one electron, which is  $-1.602 \times 10^{-19}$  coulombs. Thus,  $-1$  coulomb is the charge on  $6.24 \times 10^{18}$  electrons. The current through a specified area is defined by the electric charge passing through the area per unit of time. Thus,  $q$  is defined as the charge expressed in coulombs (C).

**Charge** is the quantity of electricity responsible for electric phenomena.

$$1\text{C} = 6.24 \times 10^{18} \text{ elektron}$$

$$1\text{e} = -1.602 \times 10^{-19} \text{ coulomb}$$

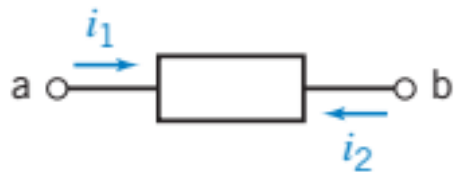


Then we can express current as

$$i = \frac{dq}{dt} \quad (1.2-1)$$

The unit of current is the ampere (A); an ampere is 1 coulomb per second.

**Current** is the time rate of flow of electric charge past a given point.

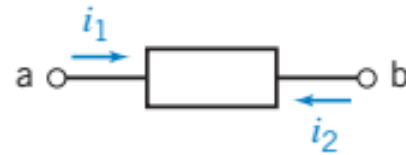


**FIGURE 1.2-3** Current in a circuit element.

They are the same size but have different directions.

$$i_1 = -i_2$$

We always associate an arrow with a current to denote its direction. A complete description of current requires both a value (which can be positive or negative) and a direction (indicated by an arrow).

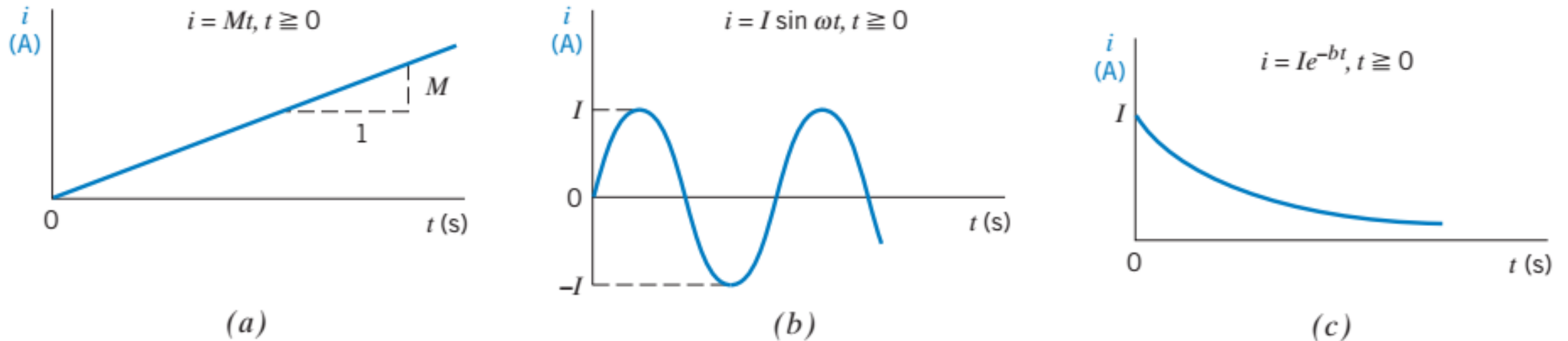


If the current flowing through an element is constant, we represent it by the constant  $I$ , as shown in Figure 1.2-4. A constant current is called a *direct current* (dc).



**FIGURE 1.2-4** A direct current of magnitude  $I$ .

A time-varying current  $i(t)$  can take many forms, such as a ramp, a sinusoid, or an exponential, as shown in Figure 1.2-5. The sinusoidal current is called an *alternating current* (ac).



**FIGURE 1.2-5** (a) A ramp with a slope  $M$ . (b) A sinusoid. (c) An exponential.  $I$  is a constant. The current  $i$  is zero for  $t < 0$ .

If the charge  $q$  is known, the current  $i$  is readily found using Eq. 1.2-1. Alternatively, if the current  $i$  is known, the charge  $q$  is readily calculated. Note that from Eq. 1.2-1, we obtain

$$q = \int_{-\infty}^t i \, d\tau = \int_0^t i \, d\tau + q(0) \quad (1.2-2)$$

where  $q(0)$  is the charge at  $t = 0$ .

### EXAMPLE 1.2-1 Current from Charge

Find the current in an element when the charge entering the element is

$$q = 12t \text{ C}$$

where  $t$  is the time in seconds.

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Find the current in an element when the charge entering the element is

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### Solution

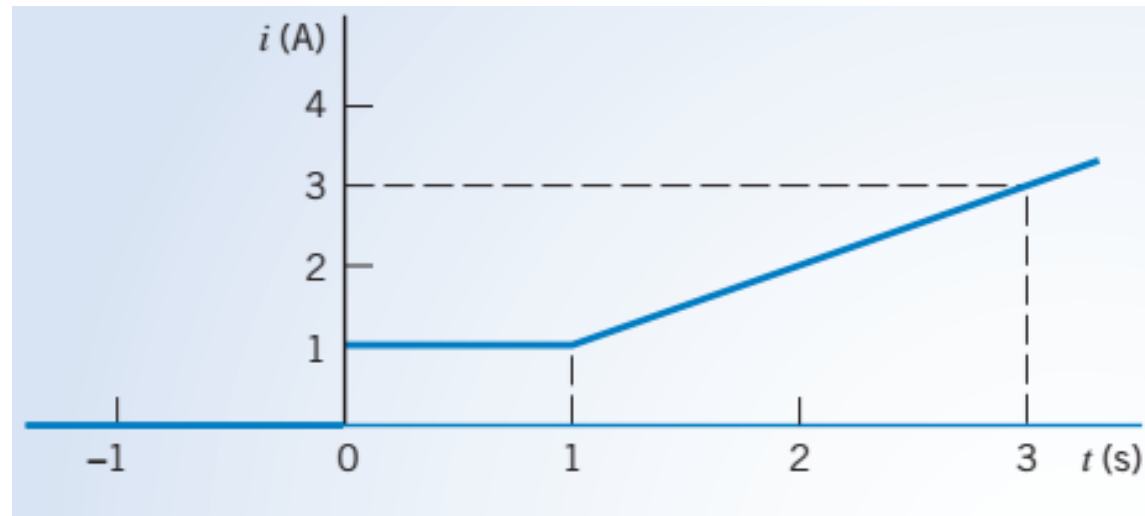
Recall that the unit of charge is coulombs, C.

$$i = \frac{dq}{dt} = 12 \text{ A}$$



**EXAMPLE 1.2-2**

Find the charge that has entered the terminal of an element from  $t=0$  s to  $t=3$  s when the current entering the element is as shown in Figure 1.2-6.



**FIGURE 1.2-6** Current waveform for Example 1.2-2.

## 1.3 *Systems of Units*

In representing a circuit and its elements, we must define a consistent system of units for the quantities occurring in the circuit. At the 1960 meeting of the General Conference of Weights and Measures, the representatives modernized the metric system and created the *Système International d'Unités*, commonly called SI units.

**SI** is *Système International d'Unités* or the International System of Units.

# 1.3 Systems of Units

**Table 1.3-1 SI Base Units**

| QUANTITY                  | SI UNIT  |        |
|---------------------------|----------|--------|
|                           | NAME     | SYMBOL |
| Length                    | meter    | m      |
| Mass                      | kilogram | kg     |
| Time                      | second   | s      |
| Electric current          | ampere   | A      |
| Thermodynamic temperature | kelvin   | K      |
| Amount of substance       | mole     | mol    |
| Luminous intensity        | candela  | cd     |

## 1.3 *Systems of Units*

**Table 1.3-2** Derived Units in SI

| QUANTITY              | UNIT NAME                   | FORMULA                        | SYMBOL   |
|-----------------------|-----------------------------|--------------------------------|----------|
| Acceleration — linear | meter per second per second | $\text{m/s}^2$                 |          |
| Velocity — linear     | meter per second            | $\text{m/s}$                   |          |
| Frequency             | hertz                       | $\text{s}^{-1}$                | Hz       |
| Force                 | newton                      | $\text{kg} \cdot \text{m/s}^2$ | N        |
| Pressure or stress    | pascal                      | $\text{N/m}^2$                 | Pa       |
| Density               | kilogram per cubic meter    | $\text{kg/m}^3$                |          |
| Energy or work        | joule                       | $\text{N} \cdot \text{m}$      | J        |
| Power                 | watt                        | $\text{J/s}$                   | W        |
| Electric charge       | coulomb                     | $\text{A} \cdot \text{s}$      | C        |
| Electric potential    | volt                        | $\text{W/A}$                   | V        |
| Electric resistance   | ohm                         | $\text{V/A}$                   | $\Omega$ |
| Electric conductance  | siemens                     | $\text{A/V}$                   | S        |
| Electric capacitance  | farad                       | $\text{C/V}$                   | F        |
| Magnetic flux         | weber                       | $\text{V} \cdot \text{s}$      | Wb       |
| Inductance            | henry                       | $\text{Wb/A}$                  | H        |

## 1.3 Systems of Units

**Table 1.3-3 SI Prefixes**

| MULTIPLE   | PREFIX | SYMBOL |
|------------|--------|--------|
| $10^{12}$  | tera   | T      |
| $10^9$     | giga   | G      |
| $10^6$     | mega   | M      |
| $10^3$     | kilo   | k      |
| $10^{-2}$  | centi  | c      |
| $10^{-3}$  | milli  | m      |
| $10^{-6}$  | micro  | $\mu$  |
| $10^{-9}$  | nano   | n      |
| $10^{-12}$ | pico   | p      |
| $10^{-15}$ | femto  | f      |

### EXAMPLE 1.3-1 SI Units

A mass of 150 grams experiences a force of 100 newtons. Find the energy or work expended if the mass moves 10 centimeters. Also, find the power if the mass completes its move in 1 millisecond.

#### Solution

The energy is found as

$$\text{energy} = \text{force} \times \text{distance} = 100 \times 0.1 = 10 \text{ J}$$

Note that we used the distance in units of meters. The power is found from

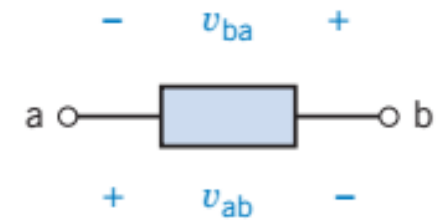
$$\text{power} = \frac{\text{energy}}{\text{time period}}$$

where the time period is  $10^{-3}$  s. Thus,

$$\text{power} = \frac{10}{10^{-3}} = 10^4 \text{ W} = 10 \text{ kW}$$

## 1.4 Voltage

The basic variables in an electrical circuit are current and voltage. These variables describe the flow of charge through the elements of a circuit and the energy required to cause charge to flow. Figure 1.4-1 shows the notation we use to describe a voltage. There are two parts to this notation: a value (perhaps represented by a variable name) and an assigned direction. The value of a voltage may be positive or negative. The direction of a voltage is given by its polarities (+, −). As a matter of vocabulary, we say that a voltage exists *across* an element. Figure 1.4-1 shows that there are two ways to label the voltage across an element. The voltage  $v_{ba}$  is proportional to the work required to move a positive charge from terminal a to terminal b. On the other hand, the voltage  $v_{ab}$  is proportional to the work required to move a positive charge from terminal b to terminal a. We sometimes read  $v_{ba}$  as “the voltage at terminal b with respect to terminal a.” Similarly,  $v_{ab}$  can be read as “the voltage at terminal a with respect to terminal b.” Alternatively, we sometimes say that  $v_{ba}$  is the voltage drop from terminal a to terminal b. The voltages  $v_{ab}$  and  $v_{ba}$  are similar but different. They have the same magnitude but different polarities. This means that



**FIGURE 1.4-1** Voltage across a circuit element.

$$v_{ab} = -v_{ba}$$

The **voltage** across an element is the work (energy) required to move a unit positive charge from the  $-$  terminal to the  $+$  terminal. The unit of voltage is the volt, V.

The equation for the voltage across the element is

$$v = \frac{dw}{dq} \quad (1.4-1)$$

where  $v$  is voltage,  $w$  is energy (or work), and  $q$  is charge. A charge of 1 coulomb delivers an energy of 1 joule as it moves through a voltage of 1 volt.



## 1.5 Power and Energy

**Power** is the time rate of supplying or receiving power.

Thus, we have the equation

The power has units of watts.

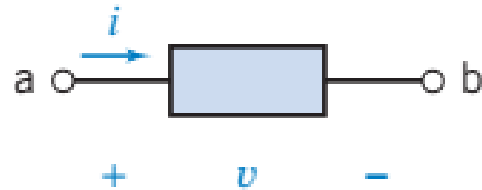
$$p = \frac{dw}{dt} \quad (1.5-1)$$

where  $p$  is power in watts,  $w$  is energy in joules, and  $t$  is time in seconds. The power associated with the current through an element is

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = v \cdot i \quad (1.5-2)$$

**Table 1.5-1 Power Received or Supplied by an Element**

POWER RECEIVED BY AN ELEMENT

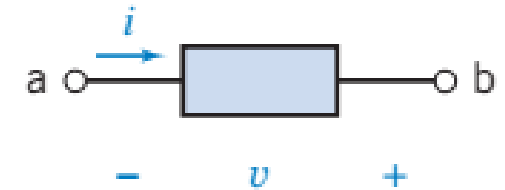


Because the reference directions of  $v$  and  $i$  adhere to the passive convention, the power

$$p = vi$$

is the power received by the element.

POWER SUPPLIED BY AN ELEMENT



Because the reference directions of  $v$  and  $i$  do not adhere to the passive convention, the power

$$p = vi$$

is the power supplied by the element.

The power received by an element and the power supplied by that same element are related by

$$\text{power received} = -\text{power supplied}$$

from Eq. 1.5-1 by rewriting it as

$$dw = p dt \quad (1.5-3)$$

On integrating, we have

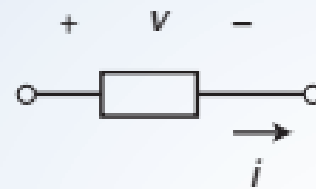
$$w = \int_{-\infty}^t p d\tau \quad (1.5-4)$$

If the element only receives power for  $t \geq t_0$  and we let  $t_0 = 0$ , then we have

$$w = \int_0^t p d\tau \quad (1.5-5)$$

### EXAMPLE 1.5-1 Electrical Power and Energy

### EXAMPLE 1.5-1 Electrical Power and Energy



**FIGURE 1.5-2** The element considered in Example 1.5-1.

Let us consider the element shown in Figure 1.5-2 when  $v = 8 \text{ V}$  and  $i = 25 \text{ mA}$ . Find the power received by the element and the energy received during a 10-ms interval.

#### Solution

In Figure 1.5-2 the current  $i$  and voltage  $v$  adhere to the passive convention. Consequently the power

$$p = vi = 8 (0.025) = 0.2 \text{ W} = 200 \text{ mW}$$

is the power *received* by the circuit element. Next, the energy received by the element is

$$w = \int_0^t p \, dt = \int_0^{0.010} 0.2 \, dt = 0.2(0.010) = 0.002 \text{ J} = 2 \text{ mJ}$$



### EXAMPLE 1.5-3 Power, Energy, and the Passive Convention

Consider the circuit shown in Figure 1.5-4 with  $v(t) = 12e^{-8t}$  V and  $i(t) = 5e^{-8t}$  A for  $t \geq 0$ . Both  $v(t)$  and  $i(t)$  are zero for  $t < 0$ . Find the power supplied by this element and the energy supplied by the element over the first 100 ms of operation.



FIGURE 1.5-4 The element considered in Example 1.5-3.

### Solution

The power

$$p(t) = v(t) i(t) = (12e^{-8t})(5e^{-8t}) = 60e^{-16t} \text{ W}$$

is the power *supplied* by the element because  $v(t)$  and  $i(t)$  do not adhere to the passive convention. This element is supplying power to the charge flowing through it.

The energy supplied during the first 100 ms = 0.1 seconds is

$$\begin{aligned} w(0.1) &= \int_0^{0.1} p \, dt = \int_0^{0.1} (60e^{-16t}) \, dt \\ &= 60 \left. \frac{e^{-16t}}{-16} \right|_0^{0.1} = -\frac{60}{16} (e^{-1.6} - 1) = 3.75(1 - e^{-1.6}) = 2.99 \text{ J} \end{aligned}$$

### EXAMPLE 1.5-4 Energy in a Thunderbolt

The average current in a typical lightning thunderbolt is  $2 \times 10^4$  A, and its typical duration is 0.1 s (Williams, 1988). The voltage between the clouds and the ground is  $5 \times 10^8$  V. Determine the total charge transmitted to the earth and the energy released.

#### Solution

The total charge is

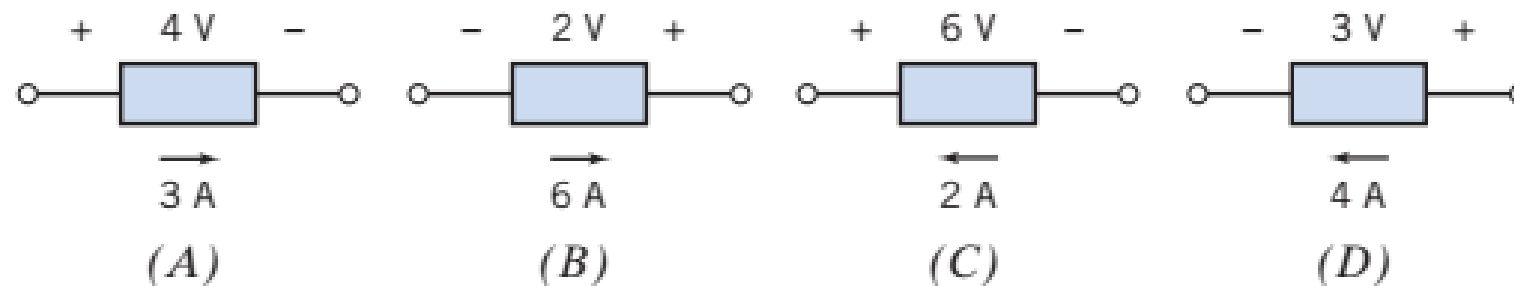
$$Q = \int_0^{0.1} i(t) dt = \int_0^{0.1} 2 \times 10^4 dt = 2 \times 10^3 \text{ C}$$

The total energy released is

$$w = \int_0^{0.1} i(t) \times v(t) dt = \int_0^{0.1} (2 \times 10^4)(5 \times 10^8) dt = 10^{12} \text{ J} = 1 \text{ TJ}$$

**EXERCISE 1.5-1** Figure E 1.5-1 shows four circuit elements identified by the letters *A*, *B*, *C*, and *D*.

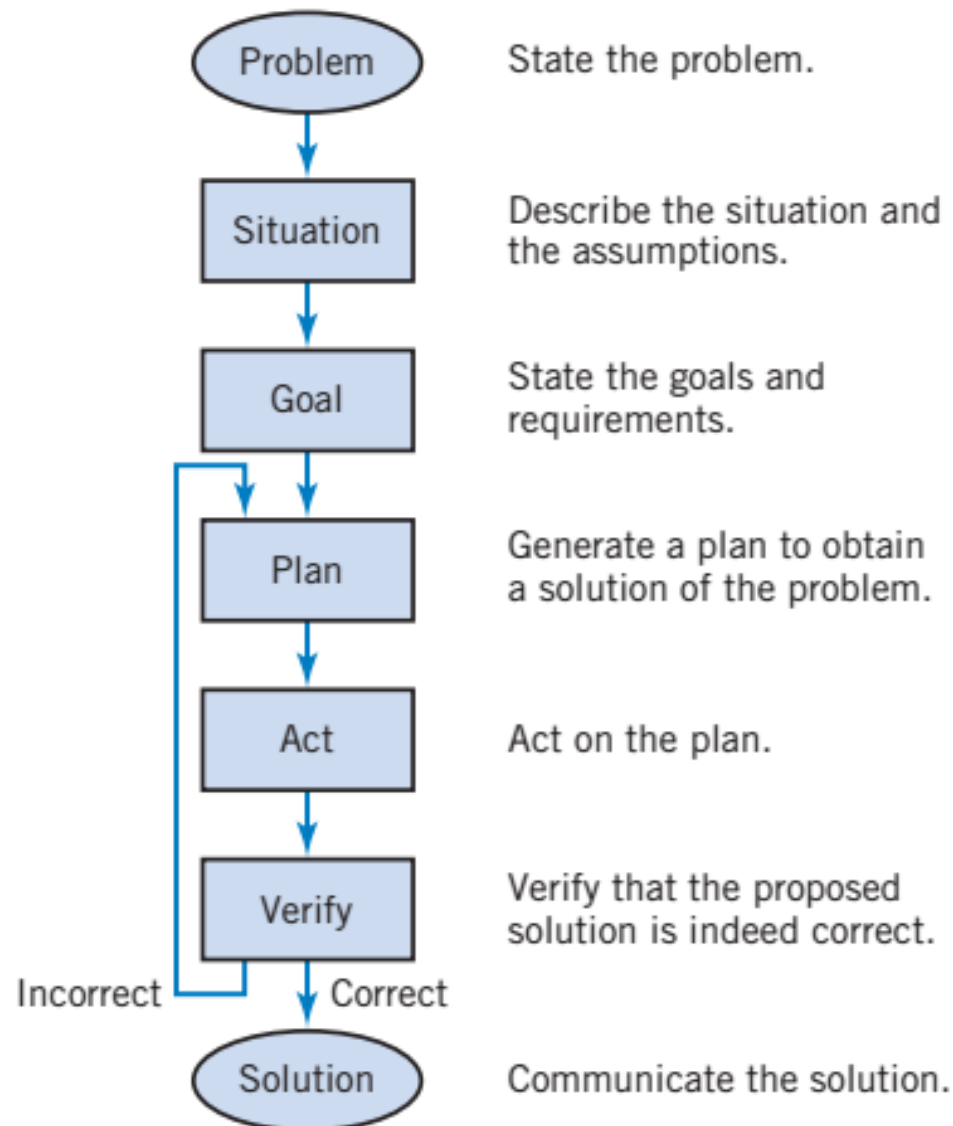
- (a) Which of the devices supply 12 W?
- (b) Which of the devices absorb 12 W?
- (c) What is the value of the power received by device *B*?
- (d) What is the value of the power delivered by device *B*?
- (e) What is the value of the power delivered by device *D*?



**FIGURE E 1.5-1**

Uygulama dersinde çözülecektir..

## 1.6 Circuit Analysis and Design



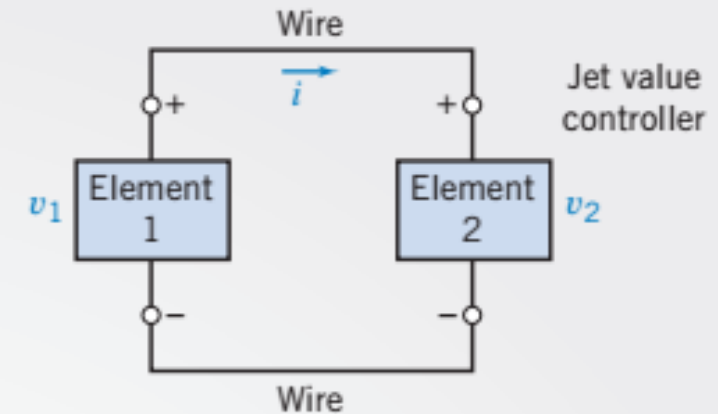
**FIGURE 1.6-1** The problem-solving method.



## 1.8 DESIGN EXAMPLE Jet Valve Controller

A small, experimental space rocket uses a two-element circuit, as shown in Figure 1.8-1, to control a jet valve from point of liftoff at  $t=0$  until expiration of the rocket after one minute. The energy that must be supplied by element 1 for the one-minute period is 40 mJ. Element 1 is a battery to be selected.

It is known that  $i(t) = De^{-t/60}$  mA for  $t \geq 0$ , and the voltage across the second element is  $v_2(t) = Be^{-t/60}$  V for  $t \geq 0$ . The maximum magnitude of the current,  $D$ , is limited to 1 mA. Determine the required constants  $D$  and  $B$  and describe the required battery.



**FIGURE 1.8-1** The circuit to control a jet valve for a space rocket.

Tasarım adımlarını düşünün..

## 1.9 SUMMARY

- Charge is the intrinsic property of matter responsible for electric phenomena. The current in a circuit element is the rate of movement of charge through the element. The voltage across an element indicates the energy available to cause charge to move through the element.
- Given the current,  $i$ , and voltage,  $v$ , of a circuit element, the power,  $p$ , and energy,  $w$ , are given by

$$p = v \cdot i \quad \text{and} \quad w = \int_0^t p d\tau$$

- Table 1.5-1 summarizes the use of the passive convention when calculating the power supplied or received by a circuit element.
- The SI units (Table 1.3-1) are used by today's engineers and scientists. Using decimal prefixes (Table 1.3-3), we may simply express electrical quantities with a wide range of magnitudes.